Learning Check

1. (True or False.) The flash from a camera is an example of a continuous wave.
2. A ________________ wave is one with the oscillations perpendicular to the direction of wave propagation.
3. The only way to make sound travel faster through air is to increase the air's ________________.
4. (True or False.) Regions in a longitudinal wave where the particles of the medium are squeezed together are called compressions.
5. Two sound waves traveling through air have different frequencies. Consequently, they must also have different
   (a) amplitudes.
   (b) speeds.
   (c) wavelengths.
   (d) All of the above.

ANSWERS: 1. False **2.** transverse **3.** temperature **4.** True **5.** (c)

Learning Check

1. As a sound wave or water ripple travels out from its source, its ________________ decreases.
2. Which of the following can change the frequency of a wave?
   (a) interference
   (b) Doppler effect
   (c) diffraction
   (d) All of the above.
3. (True or False.) The amplitude of a sound wave can be increased by making it reflect off a curved surface.
4. A sophisticated echolocation system can determine the following about an object by reflecting a wave off of it:
   (a) in what direction it is located
   (b) how far away it is
   (c) how fast it is approaching or moving away
   (d) All of the above.
5. When two identical waves undergo ________________ interference, the net amplitude is zero.

ANSWERS: 1. amplitude **2.** (b) **3.** True **4.** (d) **5.** destructive

Learning Check

1. (True or False.) Sound cannot travel through solid steel.
2. We are able to hear because our ears respond to the changes in ______________ associated with a sound wave.
3. Which of the following is *not* one of the ways sound can be used?
   (a) to make a bubble in water produce light
   (b) to relay information to orbiting satellites
   (c) to form images of internal organs
   (d) to cool small volumes to very low temperatures

ANSWERS: 1. False **2.** pressure **3.** (b)